Software Tester (T3)

Project: Automated UI Testing for an E-Commerce Website

Objective: To write a clean, maintainable, and automated suite of UI tests for the key user journeys of a public e-commerce website.
Description: Using Selenium WebDriver, you will develop automated test scripts to verify the functionality of a live e-commerce website. The project will focus on core testing skills: identifying user flows, writing scripts, adding verifications (assertions), and structuring the code for reusability.
Technologies to Use: Selenium WebDriver, Test Case Development, and a programming language like Python or Java.

Team Members

1. Mohammed Ahmed Atta (Leader)
2. Zyad Yasser Omar
3. Hafsa Nabil Yahia
4. Hanaa Mohamed Okla
5. Mariam Mohamed Farrag

Week 1

Tasks Description With Team Member Responsibility:

- **Mohammed Ahmed Atta:** Basic project structure - Create a simple folder structure to organize your test scripts.
- **Zyad Yasser Omar & Hafsa Nabil Yahia:** Set up the testing environment (Install Selenium WebDriver/ChromeDriver) and Select a target website (e.g., Sauce Labs).
- **Hanaa Mohamed Okla & Mariam Mohamed Farrag:** Write your first test - Create a single, simple automated test script that opens the website's homepage and verifies that the page title is correct.

Deliverables:

1. A working local environment where you can run Selenium tests.
2. A single, successful test script that launches a browser and performs a simple check.
3. A document identifying the target website for the project.

Week 2

Tasks Description With Team Member Responsibility:

- **Hafsa Nabil Yahia:** Develop test cases - Write simple, manual test cases for successfully logging in and searching/adding a product to the cart.
- **Zyad Yasser Omar:** Automate test cases - Write new Selenium scripts to automate the login user journey.
- **Hanaa Mohamed Okla:** Automate test cases - Write new Selenium scripts to automate the "add to cart" user journey.
- **Mariam Mohamed Farrag & Mohammed Ahmed Atta:** Implement basic waits - Use Selenium's "waits" to ensure your scripts are stable.

Deliverables:

1. An automated test script for the user login process.
2. An automated test script for searching for and adding an item to the cart.
3. Test scripts that run reliably without random timing-related failures.

Week 3

Tasks Description With Team Member Responsibility:

- **Mariam Mohamed Farrag:** Automate checkout process - Write a test script that navigates to the shopping cart, verifies items, and fills dummy info at checkout.
- **Hanaa Mohamed Okla & Zyad Yasser Omar:** Add detailed assertions - Enhance all test scripts with specific assertions to verify the application's state.
- **Mohammed Ahmed Atta:** Handle test data - Practice managing test data within your scripts, such as usernames and passwords.

Deliverables:

1. A complete, automated test script for the checkout flow.
2. All test scripts updated with clear pass/fail conditions using assertions.

Week 4

Tasks Description With Team Member Responsibility:

- **Zyad Yasser Omar & Mariam Mohamed Farrag:** Refactor with Page Object Model (POM) - Rewrite test scripts separating UI locators and test logic.
- **Hafsa Nabil Yahia & Hanaa Mohamed Okla:** Organize into a test suite and Execute the full suite.
- **Mohammed Ahmed Atta:** Write a final summary report - Create a summary of tests, features covered, and final pass/fail results.

Deliverables:

1. Fully refactored test code organized using the Page Object Model.
2. A final summary report detailing the project's test coverage and outcomes.
3. A clean, maintainable, and well-documented testing project.